

1. (a) Many good examples have been suggested (e.g., X_1 being STR while X_2 being certain demographic factor that both affects the student performance (Y) and correlates with STR).

In general, as long as we can show that X_1 (included regressor) and X_2 (omitted variable) are correlated, and X_2 has effect on Y , then we can say that there exists concern of OVB from X_2 (in practice, how serious is this concern will depend on whether X_1 and X_2 are highly correlated and also the value of the coefficient β_2).

Also, the sign of the OVB can be seen from the following table:

	$Cov(X_1, X_2) > 0$ ($Corr(X_1, X_2) > 0$)	$Cov(X_1, X_1) < 0$ ($Corr(X_1, X_2) < 0$)
$\beta_2 > 0$	Positive (upward) bias	Negative (downward) bias
$\beta_2 < 0$	Negative (downward) bias	Positive (upward) bias

Table 1: Sign of the OVB

- (b) The multicollinearity arises when one of the regressors is a perfect linear combination of the other regressors. An example of multicollinearity is the dummy variable trap, which occurs when we use G dummy variables for G categories (when we already have a constant term in the model).

For instance, define $Suburb_i = 1$ if the i -th district is in suburb area and 0 otherwise, $Urban_i = 1$ if the i -th district is in urban area and 0 otherwise, and $Rural_i = 1$ if the i -th district is in rural area and 0 otherwise.

Then, the model

$$TestScore_i = \beta_0 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i, \quad i = 1, \dots, n, \quad (1)$$

has the **dummy variable trap** because (i) the model in fact can be written as:

$$TestScore_i = \beta_0 \cdot 1 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i, \quad i = 1, \dots, n, \quad (2)$$

and (ii) for all i in the data,

$$Suburb_i + Urban_i + Rural_i = 1. \quad (3)$$

Here, 1 corresponds to the **constant term** [the implicit regressor for the intercept; therefore, here we in fact have 4 explicit regressors (STR, Subrub, Urban, and Rural) and 1 implicit regressor].

This problem can be solved easily by omitting any one of the three dummy variables. For example, if we omit $Suburb_i$, then we will have $\mathbf{Urban}_i + \mathbf{Rural}_i \neq \mathbf{1}$ for some i . This solves the problem of dummy variable trap.

In addition, notice that we cannot solve the dummy variable trap by simply changing 1 to some other number such as 2:

$$TestScore_i = \beta_0 \cdot 2 + \beta_1 STR_i + \beta_2 Suburb_i + \beta_3 Urban_i + \beta_4 Rural_i + u_i, \quad i = 1, \dots, n. \quad (4)$$

The dummy variable trap still exists because for all i in the data,

$$2(Suburb_i + Urban_i + Rural_i) = 2, \quad (5)$$

that is, there still exists a linear combination among the regressors.

Similarly, for the gender-wage-gap model discussed in class, suppose we define

$$wage_i = \beta_0 \cdot 1 + \beta_1 D_{1i} + \beta_2 D_{2i} + u_i, \quad i = 1, \dots, n$$

where

$$D_{1i} = \begin{cases} 1 & \text{if } i \text{ is male} \\ 0 & \text{if } i \text{ is female} \end{cases} \quad \text{and} \quad D_{2i} = \begin{cases} 1 & \text{if } i \text{ is female} \\ 0 & \text{if } i \text{ is male.} \end{cases}$$

This model is not estimable because of the dummy variable trap.

2. (a) The OVB is the bias in the OLS estimator that arises when the explanatory variable is correlated with a variable that is omitted from the regression. The OVB occurs if the following two conditions hold:
 - (1) The explanatory variable (regressor) is correlated with the omitted variable;
 - (2) The omitted variable is a determinant of the dependent variable (the omitted variable has an effect on the dependent variable).
- (b) The probable signs of coefficients: β_1 negative and β_2 positive.
- (c) The simple regression model with $rooms_i$ omitted can be written as:

$$price_i = \beta_0 + \beta_1 nox_i + u_i, \quad i = 1, \dots, n. \quad (6)$$

where $u_i = \beta_2 room_i + e_i$. From the formula of OVB, we have:

$$OVB = \beta_2 \frac{Cov(nox_i, rooms_i)}{Var(nox_i)}. \quad (7)$$

Note that β_2 is positive, and $Cov(nox_i, rooms_i)$ is negative because $Corr(nox_i, rooms_i)$ is negative. Therefore, the OVB is negative, which means that the simple regression of $price_i$ on nox_i produces a downward biased estimator of β_1 .

3. (a) The multiple regression model in column (1) can be written as:

$$AHE_i = \beta_0 + \beta_1 College_i + \beta_2 Female_i + u_i, \quad i = 1, \dots, n. \quad (8)$$

Then, according to the results in column (1), on average, workers with college degrees earn more, than workers with only high school diplomas by 10.47 (dollars per hour).

- (b) The multiple regression model in column (2) can be written as:

$$AHE_i = \beta_0 + \beta_1 College_i + \beta_2 Female_i + \beta_3 Age_i + u_i, \quad i = 1, \dots, n. \quad (9)$$

Then, according to the results in column (2), the prediction for Sally is:

$$\widehat{AHE} = 0.11 + 10.44 \cdot 1 - 4.56 \cdot 1 + 0.61 \cdot 29 = 23.68 \text{ (dollars per hour)}. \quad (10)$$

The prediction for Besty is:

$$\widehat{AHE} = 0.11 + 10.44 \cdot 1 - 4.56 \cdot 1 + 0.61 \cdot 34 = 26.73 \text{ (dollars per hour)}. \quad (11)$$

- (c) i. *West* is omitted from the regression to avoid the dummy variable trap. As discussed in question 1(b), if it were included there would be a perfect linear combination among the regressors (that is, *Northeast*, *Midwest*, *South*, *West* and the constant term).
- ii. The multiple regression model in column (3) can be written as:

$$\begin{aligned} AHE_i = & \beta_0 + \beta_1 College_i + \beta_2 Female_i + \beta_3 Age_i + \beta_4 Northeast_i \\ & + \beta_5 Midwest_i + \beta_6 South_i + u_i, \quad i = 1, \dots, n. \end{aligned} \quad (12)$$

Then, the prediction for Juanita is:

$$\begin{aligned} \widehat{AHE} = & 0.33 + 10.42 \cdot 1 - 4.57 \cdot 1 + 0.61 \cdot 28 + 0.74 \cdot 0 \\ & - 1.54 \cdot 0 - 0.44 \cdot 1 = 22.82 \text{ (dollars per hour)}. \end{aligned} \quad (13)$$

The prediction for Jennifer is:

$$\begin{aligned} \widehat{AHE} = & 0.33 + 10.42 \cdot 1 - 4.57 \cdot 1 + 0.61 \cdot 25 + 0.74 \cdot 0 \\ & - 1.54 \cdot 1 - 0.44 \cdot 0 = 19.89 \text{ (dollars per hour)}. \end{aligned} \quad (14)$$

Therefore, the predicted difference in earnings between the two is equal to $22.82 - 19.89 = 2.93$ (dollars per hour, which is also equal to $0.63 \cdot 3 - 0.44 + 1.54$).

4. (a) STATA command: **reg birthweight smoker, robust**

With the simple regression, the estimated effect of smoking on birth weight is -253.23 grams. (Note that “*smoker*” is a dummy variable, and is equal to 1 if the mother smokes during the pregnancy and equal to 0 otherwise.)

(b) STATA command: **reg birthweight smoker nprevist, robust**

With the multiple regression, the estimated effect of smoking on birth weight is -218.83 grams. The difference between the simple regression and multiple regression is 34.4 grams, or around 16% of the estimated result ($34.4/218.83 \approx 0.16$).

The multiple regression model is

$$birthweight_i = \beta_0 + \beta_1 smoker_i + \beta_2 nprevist_i + u_i, \quad i = 1, \dots, n. \quad (15)$$

To discuss the OVB, we first need to determine the signs of β_2 and $Cov(smoker_i, nprevist_i)$.

It is reasonable to think that number of prenatal visit will have positive effect on baby's birth weight, so β_2 is positive. In addition, note that the correlation between “*smoker*” and “*nprevist*” is -0.11 (STATA command: **correlate smoker nprevist**), so their covariance should also be negative. Therefore, we conclude that the OVB is negative by using the OVB formula, which means that the simple regression in (a) produces downward bias, and this conclusion is in fact coherent with the actual estimation result (-253.23 vs -218.83).

A natural explanation of the negative correlation between “*smoker*” and “*nprevist*” is that if a mother smokes during pregnancy, then she may be relatively careless for the new baby, so she is less likely to go to hospitals or clinics for prenatal care visits.

(c)

$$\begin{aligned} \widehat{birthweight} &= \hat{\beta}_0 + \hat{\beta}_1 smoker + \hat{\beta}_2 nprevist \\ &= 3050.53 - 218.83 \cdot 1 + 34.1 \cdot 8 \\ &= 3104.5 \text{ (grams)} \end{aligned} \quad (16)$$